Seventy-Five Seats

84-355 Democracy's Data | Population and House seats by region, 1910–2020

Your name here

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Everything here runs as given. Your job is to read what comes out and say what it means. Base R only, no packages.

> **Read `regional-shift-brief.Rmd` first.** This lab runs the same numbered steps with the code exposed. The brief tells you what the question is; this shows you where each number came from.
>
> **And read `apportionment` first if you have not.** That lab takes a *single* reapportionment apart — Huntington–Hill, and New York missing the 435th seat by 89 people. **This lab is about sixty years of them.** The arithmetic is assumed here, not repeated.

Where did that number come from?

> *"Texas gained two seats and New York lost one."*

That is a real headline from April 2021, and it is accurate. It is also the smallest true thing you can say about what happened.

The United States redistributes political power every ten years without anybody voting on it. People move for jobs and weather; the census counts them; 435 seats and 538 electoral votes follow. **The question in this lab is what that adds up to when you stop looking one decade at a time.**

Steps 1–3 — One record, the file, and why we start at 1960

```
st  <- read.csv("data/states.csv",     stringsAsFactors = FALSE)
reg <- read.csv("data/regions.csv",    stringsAsFactors = FALSE)
nat <- read.csv("data/nation.csv",     stringsAsFactors = FALSE)
sc  <- read.csv("data/seatchange.csv", stringsAsFactors = FALSE)
sd  <- read.csv("data/southdefs.csv",  stringsAsFactors = FALSE)

c(state_years = nrow(st), states = length(unique(st$name)),
  first = min(st$year), last = max(st$year), censuses = length(unique(st$year)))
```

```
#> state_years      states       first        last    censuses
#>         612          51        1910        2020          12
```

One row per state per census. Here is one state, twice:

```r
st[st$name == "Florida" & st$year %in% c(1960, 2020),
   c("name", "year", "pop", "reps", "repchg", "region")]
```

Name	Year	Pop	Reps	Repchg	Region
Florida	1960	4951560	12	4	South
Florida	2020	21538187	28	1	South

Three things the build script had to survive, all documented in `data/build-data.R`:

1. Every number in the source file is a **quoted string with commas** — `"4,951,560"`. `as.numeric()` returns `NA` for all of them, silently.
2. **Puerto Rico is a `State` row and is not in the national total.** The **District of Columbia** is in the national total and has no representative.
3. Alaska and Hawaii are in the file from 1910 with **no representatives**:

```r
st[st$name %in% c("Alaska", "Hawaii") & st$year %in% c(1950, 1960),
   c("name", "year", "pop", "reps")]
```

Name	Year	Pop	Reps
Alaska	1950	128643	NA
Alaska	1960	226167	1
Hawaii	1950	499794	NA
Hawaii	1960	632772	2

They were territories. **The fifty-state House begins in 1960**, which is why every seat figure below runs 1960–2020. Population runs back to 1910 and is labelled as such.

One more thing, and it is unusual enough to be worth a line. The Census Bureau publishes its **own** region totals in the same file, so the state-to-region mapping used here is checkable rather than assumed:

```r
mine <- aggregate(pop ~ region + year, st, sum)
c(region_years_checked = nrow(mine),
  national_totals_match = all(tapply(st$pop, st$year, sum) ==
                              nat$pop[order(nat$year)]))
```

```
#>  region_years_checked national_totals_match
#>                    48                     1
```

`build-data.R` reruns that comparison and **stops** if any region-year disagrees. That is the only real way to know a lookup table is right.

Step 4 — One decade at a time

```r
d20 <- st[st$year == 2020 & !is.na(st$repchg) & st$repchg != 0,
          c("name", "region", "reps", "repchg")]
d20
```

Name	Region	Reps	Repchg
California	West	52	-1
Colorado	West	8	1
Florida	South	28	1
Illinois	Midwest	17	-1
Michigan	Midwest	13	-1
Montana	West	2	1
New York	Northeast	26	-1
North Carolina	South	14	1
Ohio	Midwest	15	-1
Oregon	West	6	1
Pennsylvania	Northeast	17	-1
Texas	South	38	2
West Virginia	South	2	-1

```r
c(seats_that_moved = sum(d20$repchg[d20$repchg > 0]),
  states_affected  = nrow(d20),
  largest_single_move = max(abs(d20$repchg)))
```

```
#>    seats_that_moved     states_affected largest_single_move
#>                   7                  13                   2
```

Seven seats, thirteen states, none moving more than two. And that is typical:

```r
dec <- st[st$year > 1960 & st$name != "District of Columbia" & !is.na(st$repchg), ]
moved_by_decade <- tapply(dec$repchg, dec$year, function(x) sum(x[x > 0]))

rbind(seats_changing_states = moved_by_decade,
      states_affected       = tapply(dec$repchg, dec$year, function(x) sum(x != 0)),
      pct_of_the_house      = round(100 * moved_by_decade / 435, 1))
```

```
#>                       1970 1980 1990 2000 2010 2020
#> seats_changing_states 11.0 17.0 19.0 12.0 12.0  7.0
#> states_affected       14.0 21.0 21.0 18.0 18.0 13.0
#> pct_of_the_house       2.5  3.9  4.4  2.8  2.8  1.6
```

Under five percent of the House even in the busiest decade. Taken alone, each reapportionment reads as maintenance.

> **Stop here and write down two numbers before you run the next chunk.**
>
> New York's worst single decade in this period was **5 seats**, and most decades it lost one or two. **How many seats do you think New York lost in total between 1960 and 2020?**
>
> And, adding up the top row above: **how many seats do you think have changed states altogether?**

Step 5 — The pivot: sixty years at once

```r
c(seats_that_changed_states = sum(sc$change[sc$change > 0]),
  pct_of_the_house = round(100 * sum(sc$change[sc$change > 0]) / 435, 1),
  states_that_gained = sum(sc$change > 0),
  states_that_lost   = sum(sc$change < 0),
  states_unchanged   = sum(sc$change == 0))
```

```
#> seats_that_changed_states          pct_of_the_house        states_that_gained
#>                      75.0                      17.2                      14.0
#>          states_that_lost          states_unchanged
#>                      21.0                      15.0
```

```r
sc[order(-abs(sc$change)), c("name", "region", "reps_1960", "reps_2020", "change")][1:10, ]
```

Name	Region	Reps 1960	Reps 2020	Change
Florida	South	12	28	16
Texas	South	23	38	15
New York	Northeast	41	26	-15
California	West	38	52	14
Pennsylvania	Northeast	27	17	-10
Ohio	Midwest	24	15	-9
Illinois	Midwest	24	17	-7
Arizona	West	3	9	6
Michigan	Midwest	19	13	-6
Georgia	South	10	14	4

Seventy-five seats. Seventeen percent of the House of Representatives sits in a different state than it did in 1960, and no single decade ever announced it.

New York lost **15**. Put that next to the size of everybody else's delegation:

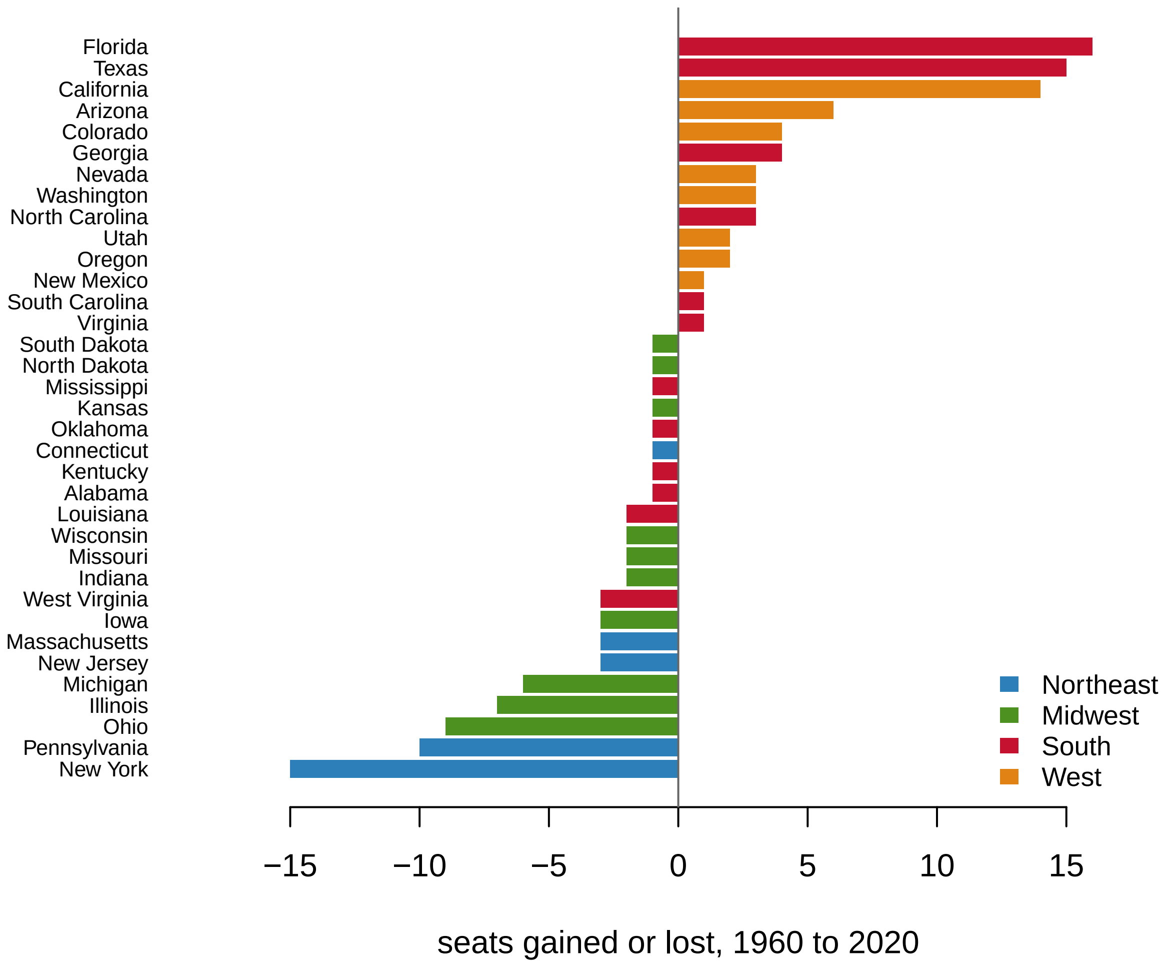
```r
ny <- sc[sc$name == "New York", ]
c(new_york_1960 = ny$reps_1960, new_york_2020 = ny$reps_2020,
  lost = abs(ny$change),
  states_smaller_than_that_loss = sum(sc$reps_2020 <= abs(ny$change)))
```

```
#>                 new_york_1960                 new_york_2020
#>                            41                            26
#>                          lost states_smaller_than_that_loss
#>                            15                            44
```

44 of the 50 states do not have 15 seats in total.

```r
m <- sc[sc$change != 0, ]; m <- m[order(m$change), ]
cols <- c(Northeast = "#2c7fb8", Midwest = "#4d9221",
          South = "#C41230", West = "#e08214")

par(mar = c(4.2, 8.4, 1, 2))
```

```
barplot(m$change, horiz = TRUE, las = 1, names.arg = m$name,
        col = cols[m$region], border = NA, cex.names = 0.7, xlim = c(-19, 19),
        xlab = "seats gained or lost, 1960 to 2020")
abline(v = 0, col = "#666")
legend("bottomright", names(cols), fill = cols, border = NA, bty = "n", cex = 0.8)
```



Step 6 — Regions, and whether the movement ever reverses

```
blk <- ifelse(sc$region %in% c("Northeast", "Midwest"), "NE+MW", "S+W")

rbind(seats_1960 = tapply(sc$reps_1960, sc$region, sum),
      seats_2020 = tapply(sc$reps_2020, sc$region, sum),
      change     = tapply(sc$change,    sc$region, sum))
```

```
#>            Midwest Northeast South West
#> seats_1960     125       108   133   69
#> seats_2020      91        76   164  104
#> change         -34       -32    31   35
```

```r
rbind(seats_1960 = tapply(sc$reps_1960, blk, sum),
      seats_2020 = tapply(sc$reps_2020, blk, sum),
      pct_2020   = round(100 * tapply(sc$reps_2020, blk, sum) / 435, 1))
```

```
#>            NE+MW   S+W
#> seats_1960 233.0 202.0
#> seats_2020 167.0 268.0
#> pct_2020    38.4  61.6
```

```r
# of the seats that changed states, how many left their region altogether?
netreg <- tapply(sc$change, sc$region, sum)
c(seats_that_changed_states = sum(sc$change[sc$change > 0]),
  crossed_a_regional_border = sum(netreg[netreg > 0]),
  moved_within_one_region   = sum(sc$change[sc$change > 0]) - sum(netreg[netreg > 0]))
```

```
#> seats_that_changed_states crossed_a_regional_border   moved_within_one_region
#>                        75                        66                         9
```

In 1960 the Northeast and Midwest together held a majority of the House — 233 of 435 seats. They now hold 167, or 38.4%. And 66 of the 75 seats that moved at all left their region entirely — this is not states trading with neighbours.

Now: is this a trend or a fluctuation? Compare the sum of the per-decade columns in Step 4 with the cumulative total in Step 5, then check each state's path.

```r
z  <- st[st$year >= 1960 & st$name != "District of Columbia", ]
sp <- split(z[order(z$year), ], z$name[order(z$year)])
mono <- sapply(sp, function(x) { d <- diff(x$reps[order(x$year)])
                                 all(d >= 0) || all(d <= 0) })

c(seat_gains_summed_by_decade = sum(dec$repchg[dec$repchg > 0]),
  net_gains_1960_to_2020      = sum(sc$change[sc$change > 0]),
  states_that_only_went_one_way = sum(mono))
```

```
#>   seat_gains_summed_by_decade       net_gains_1960_to_2020
#>                            78                           75
#> states_that_only_went_one_way
#>                            47
```

```r
names(mono)[!mono]
```

```
#> [1] "California" "Montana"    "Tennessee"
```

47 of 50 states moved in one direction only. This is not a cycle. It is a ratchet.

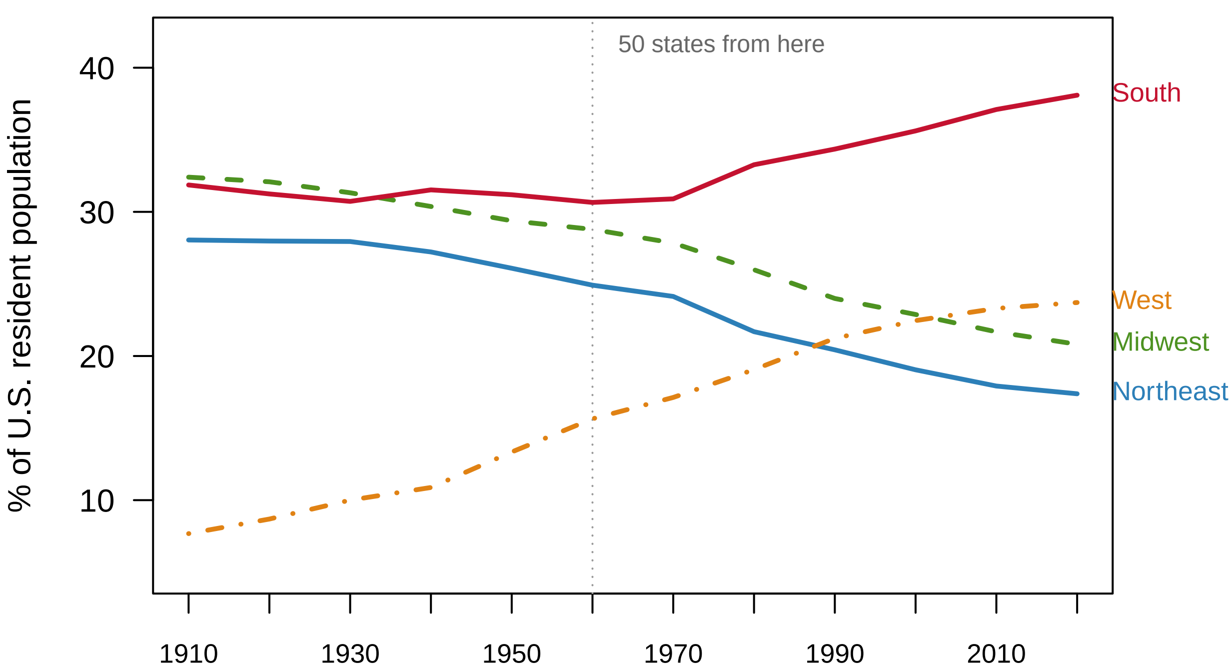
Step 7 — The long line

Seats are lumpy; a state has to gain a whole one. Population is not.

```r
wide <- reshape(reg[, c("region", "year", "share")], idvar = "year",
                timevar = "region", direction = "wide")
round(wide[order(wide$year), ], 1)
```

Year	Share.Midwest	Share.Northeast	Share.South	Share.West
1910	32.4	28.0	31.9	7.7
1920	32.1	28.0	31.2	8.7
1930	31.3	27.9	30.7	10.0
1940	30.4	27.2	31.5	10.9
1950	29.4	26.1	31.2	13.3
1960	28.8	24.9	30.7	15.6
1970	27.8	24.1	30.9	17.1
1980	26.0	21.7	33.3	19.1
1990	24.0	20.4	34.4	21.2
2000	22.9	19.0	35.6	22.5
2010	21.7	17.9	37.1	23.3
2020	20.8	17.4	38.1	23.7

```r
par(mar = c(4, 4.4, 1, 6.6))
plot(NA, xlim = c(1910, 2020), ylim = c(5, 42), las = 1, xaxt = "n",
     xlab = "", ylab = "% of U.S. resident population")
axis(1, at = seq(1910, 2020, 10), cex.axis = 0.85)
abline(v = 1960, col = "#999", lty = 3)
text(1960, 41.5, " 50 states from here", col = "#666", cex = 0.75, pos = 4)
ltys <- c(Northeast = 1, Midwest = 2, South = 1, West = 4)
for (r in names(cols)) {
  q <- reg[reg$region == r, ]; q <- q[order(q$year), ]
  lines(q$year, q$share, col = cols[r], lwd = 2.4, lty = ltys[r])
  text(2021, q$share[q$year == 2020], paste0(" ", r), col = cols[r],
       cex = 0.8, pos = 4, xpd = NA)
}
```



Four things in that picture.

- **The Northeast falls the whole way**, 28.0% to 17.4%, with no reversal.
- **The West passes the Northeast in 1990** and the Midwest in 2010.
- **The South passes the Midwest in 1940** and never gives the lead back.
- **And the South's rise starts after 1960, not before.** Between 1910 and 1960 the Census South's share of the country *fell*.

Step 8 — Does the finding depend on what "the South" means?

`house-competition` and `historical-campaigns` define the South as the eleven former Confederate states. The Census Bureau's South is sixteen states plus DC, adding Maryland, Delaware, Oklahoma, West Virginia and Kentucky.

```
o <- sd[sd$year %in% c(1960, 2020), ]
w <- reshape(o[, c("definition", "year", "share")], idvar = "definition",
             timevar = "year", direction = "wide")
w$change <- round(w$share.2020 - w$share.1960, 1)
w$share.1960 <- round(w$share.1960, 1); w$share.2020 <- round(w$share.2020, 1)
w
```

Definition	Share.1960	Share.2020	Change
Border South only (DC, DE, KY, MD, OK, WV)	6.4	5.5	-1.0
Census South (16 states + DC)	30.7	38.1	7.4
Confederate South (11 states)	24.2	32.6	8.4
Northeast	24.9	17.4	-7.5
Sun Belt (South + West)	46.3	61.8	15.5

The narrower definition gives the larger rise, which is only possible if the extra states are a drag — and they are. The border states went from 6.4% of the country to 5.5%. **So the finding is not an artefact of the definition, and the broad version understates it.**

Step 9 — What a seat is worth now

The House has been fixed at 435 by statute since 1929. The country was not.

```
nat[, c("year", "pop", "reps", "pop_per_rep")]
```

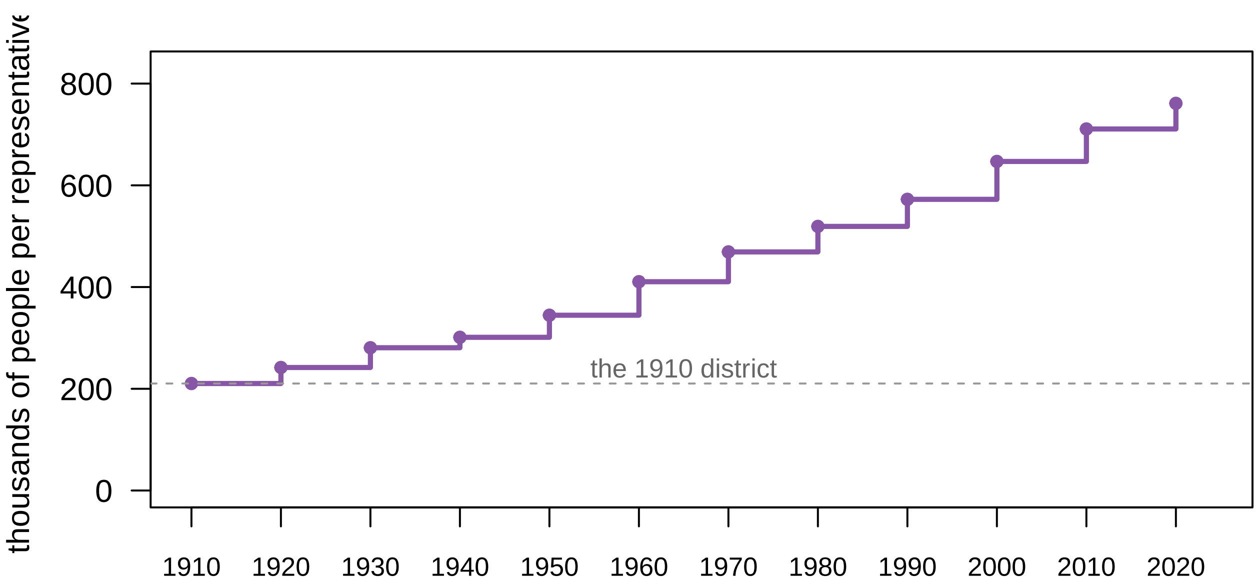
Year	Pop	Reps	Pop per rep
1910	92228531	433	210328
1920	106021568	435	241864
1930	123202660	435	280675
1940	132165129	435	301164
1950	151325798	435	344587
1960	179323175	435	410481
1970	203211926	435	469088
1980	226545805	435	519235

Year	Pop	Reps	Pop per rep
1990	248709873	435	572466
2000	281421906	435	646952
2010	308745538	435	710767
2020	331449281	435	761169

```
c(per_rep_1910 = nat$pop_per_rep[nat$year == 1910],
  per_rep_2020 = nat$pop_per_rep[nat$year == 2020],
  multiple     = round(nat$pop_per_rep[nat$year == 2020] /
                       nat$pop_per_rep[nat$year == 1910], 2),
  population_multiple = round(nat$pop[nat$year == 2020] /
                              nat$pop[nat$year == 1910], 2))
```

```
#>        per_rep_1910        per_rep_2020            multiple population_multiple
#>           210328.00           761169.00                3.62                3.59
```

```
par(mar = c(4, 5, 1, 2))
plot(nat$year, nat$pop_per_rep / 1000, type = "s", lwd = 2.6, col = "#8856a7",
     xlim = c(1910, 2024), ylim = c(0, 830), las = 1, xaxt = "n", xlab = "",
     ylab = "thousands of people per representative")
axis(1, at = seq(1910, 2020, 10), cex.axis = 0.85)
points(nat$year, nat$pop_per_rep / 1000, pch = 19, cex = 0.8, col = "#8856a7")
abline(h = nat$pop_per_rep[nat$year == 1910] / 1000, col = "#999", lty = 2)
text(1965, nat$pop_per_rep[nat$year == 1910] / 1000 + 30, "the 1910 district",
     col = "#666", cex = 0.8)
```



A member of the House now speaks for 3.62 times as many people as in 1910, and the country grew 3.59 times. Those two multiples being nearly identical *is* the mechanism: the denominator was frozen in 1929.

But bigger districts and *less equal* districts are different complaints:

```r
ppr <- st[st$year >= 1960 & st$name != "District of Columbia" & !is.na(st$reps), ]
ppr$per <- ppr$pop / ppr$reps
do.call(rbind, lapply(split(ppr, ppr$year), function(x)
  data.frame(year = x$year[1],
             smallest = x$name[which.min(x$per)],
             largest  = x$name[which.max(x$per)],
             ratio    = round(max(x$per) / min(x$per), 2))))
```

Year	Smallest	Largest	Ratio
1960	Alaska	Maine	2.14
1970	Alaska	North Dakota	2.06
1980	Montana	South Dakota	1.76
1990	Wyoming	Montana	1.76
2000	Wyoming	Montana	1.83
2010	Rhode Island	Montana	1.88
2020	Montana	Delaware	1.83

Interstate inequality got slightly better while district size nearly doubled.

Step 10 — Electoral votes moved, and nobody voted

A state's electors equal its House seats plus two.

```r
ev60 <- tapply(sc$reps_1960, blk, sum) + 2 * table(blk)
ev20 <- tapply(sc$reps_2020, blk, sum) + 2 * table(blk)
rbind(electoral_votes_1960 = ev60, electoral_votes_2020 = ev20,
      pct_1960 = round(100 * ev60 / sum(ev60), 1),
      pct_2020 = round(100 * ev20 / sum(ev20), 1))
```

```
#>                      NE+MW   S+W
#> electoral_votes_1960 275.0 260.0
#> electoral_votes_2020 209.0 326.0
#> pct_1960              51.4  48.6
#> pct_2020              39.1  60.9
```

In 1960 the Northeast and Midwest controlled a majority of the electoral votes — 275 of the 535 cast by the fifty states. By 2020 they controlled 39.1%.

DC is excluded from both columns: it had none in 1960 and got three by constitutional amendment in 1961, which is a rule change rather than a population change. Including them lowers both shares slightly and changes nothing.

Step 11 — What you can now say

You can say that 75 House seats changed states between 1960 and 2020, that 66 of them crossed a regional boundary, and that the Northeast and Midwest went from a majority of the House to 38.4% of it — with nothing enacted, proposed or voted on.

You can say it more sharply: no single decade did this. The largest reapportionment moved 19 seats and looked like housekeeping at the time.

You cannot say why people moved. This file has population and seats and nothing else. **The `labs/migration` lab measures the moving; this one shows what the moving does.**

Your turn

Change **one** thing, re-run, and write about what changed. Pick one.

Option A — Move the window

```
START <- 1960          # <- change this. Try 1970, 1980, 1990, 2000.
END   <- 2020

a <- st[st$year == START & st$name != "District of Columbia", c("name", "region", "reps")]
b <- st[st$year == END   & st$name != "District of Columbia", c("name", "reps")]
w <- merge(a, b, by = "name", suffixes = c("_start", "_end"))
w$change <- w$reps_end - w$reps_start

c(window = paste(START, "to", END), decades = (END - START) / 10,
  seats_that_changed_states = sum(w$change[w$change > 0]))
```

```
#>                    window                   decades seats_that_changed_states
#>             "1960 to 2020"                       "6"                      "75"
```

```
tapply(w$change, w$region, sum)
```

```
#>   Midwest Northeast     South      West
#>       -34       -32        31        35
```

```
head(w[order(-abs(w$change)), ], 6)
```

Name	Region	Reps start	Reps end	Change
Florida	South	12	28	16
New York	Northeast	41	26	-15
Texas	South	23	38	15
California	West	38	52	14
Pennsylvania	Northeast	27	17	-10
Ohio	Midwest	24	15	-9

Write about: does the total scale with the length of the window, or faster or slower? Compare 1960–2020 with 1990–2020 — a window half as long. If the number is more than half, the shift is accelerating; if less, it is slowing. Say which, and then say honestly how confident three or six observations let you be. Does the *ranking* of gainers and losers change, or only the size?

Option B — Redefine the South

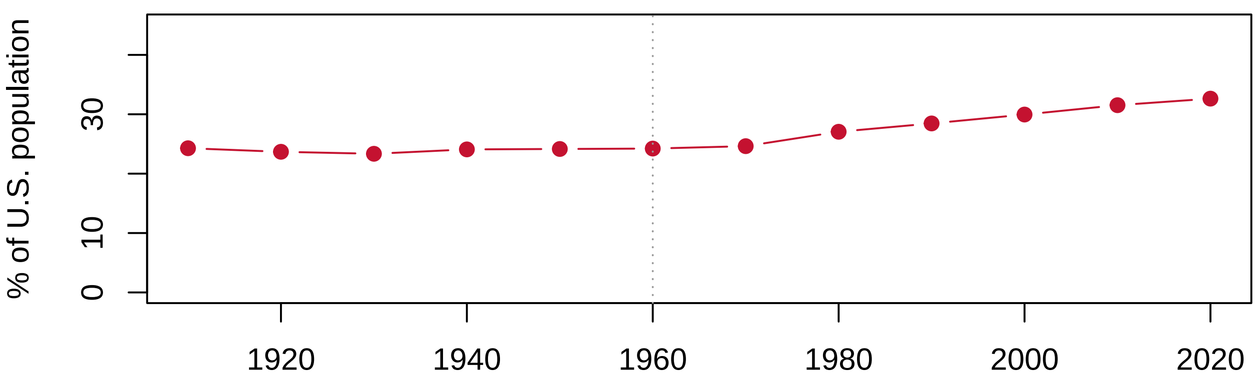
```
MY_SOUTH <- c("Alabama", "Arkansas", "Florida", "Georgia", "Louisiana",
              "Mississippi", "North Carolina", "South Carolina", "Tennessee",
              "Texas", "Virginia")     # <- add or remove states here

sel <- st[st$name %in% MY_SOUTH, ]
tot <- tapply(st$pop, st$year, sum)
share <- 100 * tapply(sel$pop, sel$year, sum) / tot

round(share, 1)
```

```
#> 1910 1920 1930 1940 1950 1960 1970 1980 1990 2000 2010 2020
#> 24.3 23.7 23.3 24.1 24.2 24.2 24.6 27.1 28.5 29.9 31.5 32.6
```

```
c(n_states = length(MY_SOUTH),
  change_1960_to_2020 = round(share["2020"] - share["1960"], 2),
  change_1910_to_1960 = round(share["1960"] - share["1910"], 2))
```

```
#>                n_states change_1960_to_2020.2020 change_1910_to_1960.1960
#>                   11.00                     8.41                    -0.06
```

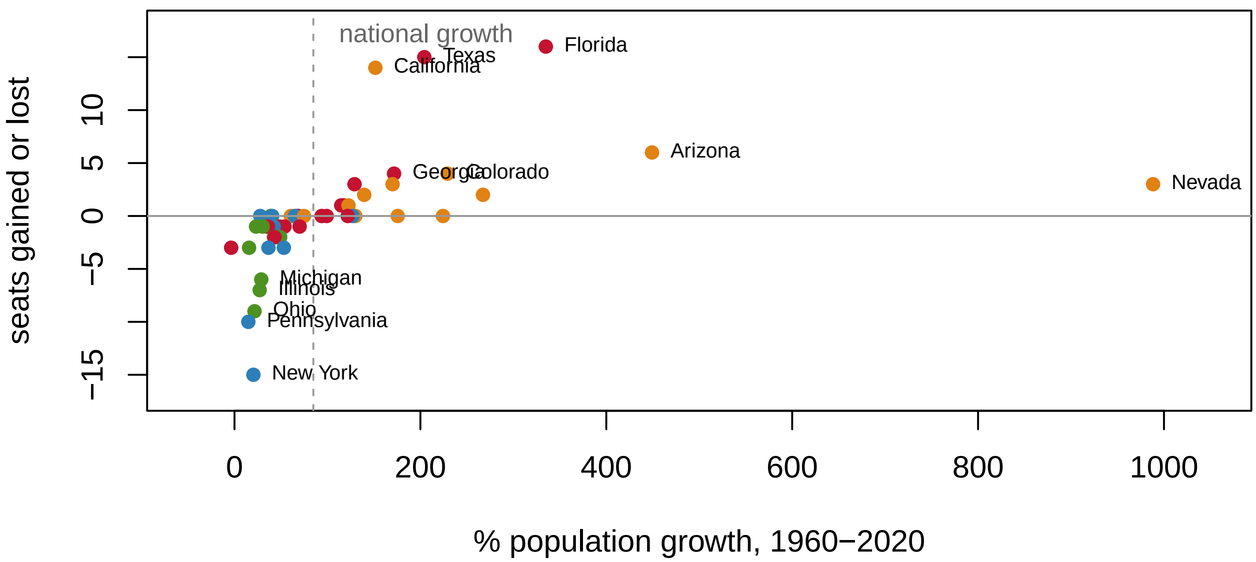
```
plot(as.numeric(names(share)), share, type = "b", pch = 19, col = "#C41230",
     ylim = c(0, 45), xlab = "", ylab = "% of U.S. population")
abline(v = 1960, lty = 3, col = "#999")
```



Write about: try adding Maryland, then Oklahoma, then Kentucky and West Virginia, then DC. Which additions make the rise *smaller*? Those states are being called Southern by the Census Bureau and are not behaving that way. Then the harder question: **the variable encodes an argument.** What claim about what the South *is* does the eleven-state list make, and what different claim does the Census list make? Which one would you defend, and does the finding survive either way?

Option C — Growth is not seats

```r
nat_growth <- 100 * (nat$pop[nat$year == 2020] / nat$pop[nat$year == 1960] - 1)

plot(sc$pop_growth, sc$change, pch = 19, col = cols[sc$region], cex = 0.9,
     xlab = "% population growth, 1960-2020", ylab = "seats gained or lost",
     xlim = c(-50, 1050), ylim = c(-17, 18))
abline(h = 0, col = "#999"); abline(v = nat_growth, lty = 2, col = "#999")
text(nat_growth, 17, " national growth", pos = 4, cex = 0.8, col = "#666")
big <- sc[abs(sc$change) >= 4 | sc$pop_growth > 400, ]
text(big$pop_growth, big$change, big$name, pos = 4, cex = 0.7)
```




```r
c(correlation = round(cor(sc$pop_growth, sc$change), 2),
  national_growth_pct = round(nat_growth, 1))
```

```
#>         correlation national_growth_pct
#>                0.45               84.80
```

```r
g <- sc[, c("name", "region", "pop_growth", "reps_1960", "reps_2020", "change")]
g$pop_growth <- round(g$pop_growth)

# the two states the paragraph below is about
g[g$name %in% c("Nevada", "Florida"), ]
```

Name	Region	Pop growth	Reps 1960	Reps 2020	Change
Florida	South	335	12	28	16
Nevada	West	988	1	4	3

```r
# states that beat the national growth rate and gained nothing
g[sc$pop_growth > nat_growth & sc$change <= 0, ]
```

Name	Region	Pop growth	Reps 1960	Reps 2020	Change
Tennessee	South	94	9	9	0
Maryland	South	99	8	8	0
Hawaii	West	130	2	2	0
Idaho	West	176	2	2	0
New Hampshire	Northeast	127	2	2	0
Alaska	West	224	1	1	0
Delaware	South	122	1	1	0

Write about: Nevada's population grew **988%** and it gained 3 seats. Florida grew 335% and gained 16. The correlation between growth and seats is only 0.45. Explain why — two reasons, and both are in the plot. Then look at the seven states that grew faster than the country and gained nothing at all, and say what they have in common. What does this tell you about the difference between *growing* and *growing faster than everybody else*?

Option D — Ask your own question

612 state-years, 12 censuses, 110 years. The 1920s are worth a look: Congress refused to reapportion at all after the 1920 census, the only time that has happened, and this file shows you the country that produced the refusal.

What to submit. Knit to HTML or PDF and upload to Canvas, with your chosen chunk and **two or three paragraphs** of interpretation.

The writing is the assignment. I am not grading your code — it is my code.

What this lab cannot tell you

- **Why anyone moved.** No jobs, wages, climate, housing costs, births, deaths or immigration. This file cannot distinguish someone arriving from Michigan from someone arriving from abroad, or either from families being larger.
- **Anything between censuses.** Twelve observations, ten years apart. A hurricane, a plant closure or a pandemic is invisible unless it survives to the next April.
- **Anything partisan.** No election results, no party of any member. The electoral-vote figures in Step 10 are arithmetic about states, not about who won.
- **How accurate the count is.** Resident population is printed as an exact integer. The Bureau publishes coverage error separately and it is not in this file. The margins here are far larger than plausible error — 15 seats is not an error bar — but `apportionment` shows a case where it decides the answer.
- **What a different rule would do.** Counting citizens only, or counting incarcerated people at their home address, would change these numbers. Neither can be computed here.

On the discussion board

By **Friday, 11:59 p.m.** A sentence or two is enough.

Three that would make good posts:

- Seventy-five seats moved between states and nobody voted on any of it. Is that a feature of the constitutional design or a problem with it?
- A representative today speaks for 3.6 times as many people as in 1910. What, if anything, does that do to representation — and why has no Congress fixed it?
- The Census Bureau's "South" includes Maryland and Oklahoma. Whose definition of a region should a federal statistical agency use, and what happens to a finding when the category is contested?

Where the data came from

U.S. Census Bureau, apportionment time series 1910–2020 — a single 40 KB CSV, no key required (https://www2.census.gov/programs-surveys/decennial/2020/data/apportionment/apportionment.csv), listed at https://www.census.gov/data/tables/time-series/dec/apportionment-data-text.html.

Three things `data/build-data.R` had to handle and documents. **Numbers arrive as quoted strings with thousands separators** and become `NA` if you convert them without stripping the commas. **Puerto Rico is a `State` row that is not in the national total, and DC is in the national total with no representative.** And **Alaska and Hawaii have no representatives before 1960**, which is why the seat window starts there.

The build script does one thing worth copying: the Bureau publishes its own four-region totals in the same file, so the script **rebuilds those rows from the state rows and stops if a single region-year disagrees.** A lookup table you wrote by hand should always be checked against something you did not.

Regions and divisions follow the Census Bureau's *Statistical Groupings of States and Counties* (Geographic Areas Reference Manual, ch. 6).